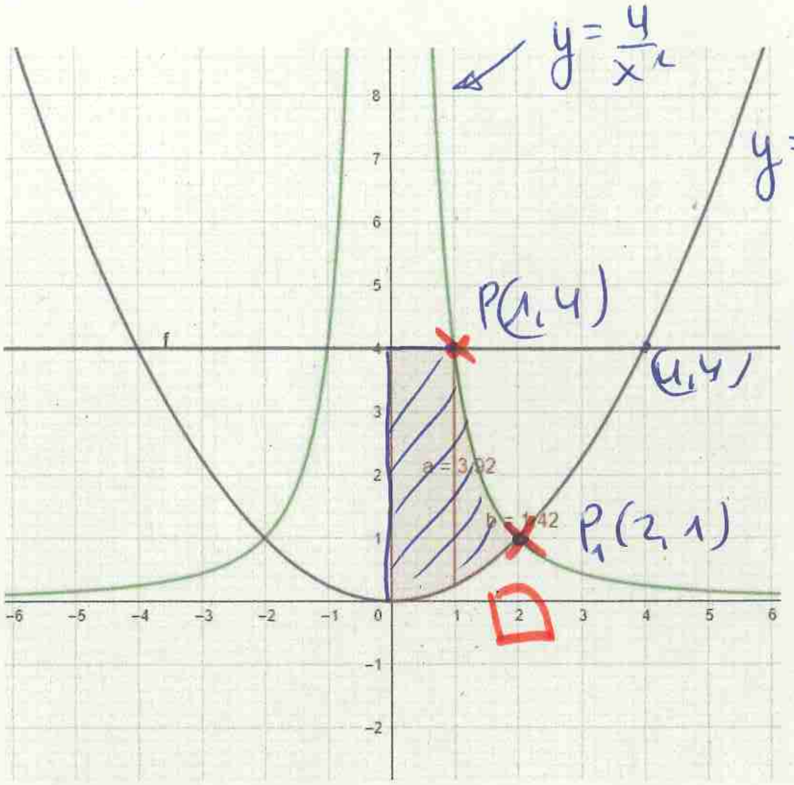
A4

A4

$y=\frac{x^4}{4}$ $\qquad y=\frac{4}{x^2}$ $\qquad y=4$



Ebaketa puntuak

• $\left.\begin{array}{l} y=\frac{4}{x^2} \\ y=\frac{x^2}{4} \end{array}\right\}$ $\frac{4}{x^2}=\frac{x^2}{4} \rightarrow x^4=16$

$x_1=2 \rightarrow y_1=1 \qquad P_1(2,1)$

$x_2=-2 \rightarrow y_2=1 \qquad P_2(-2,1)$

Lehenengo koadrantean $\boxed{P_1(2,1)}$

• $\left.\begin{array}{l} y=\frac{4}{x^2} \\ y=4 \end{array}\right\}$ $4=\frac{4}{x^2} \rightarrow x_1=1 \rightarrow y_1=4$

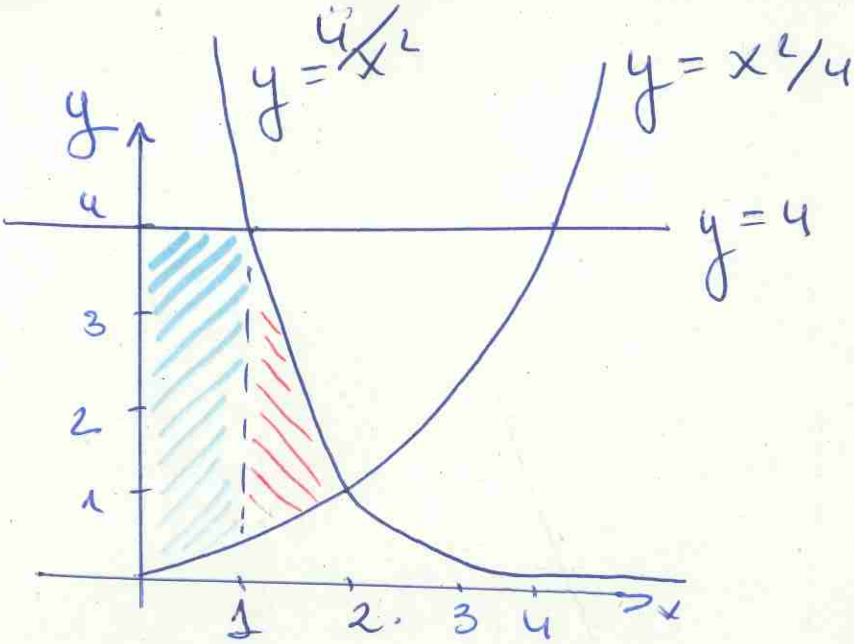
$x_2=-1 \rightarrow y_2=4$

lehenengo koadrantean $\boxed{P_2(1,4)}$

• $\left.\begin{array}{l} y=\frac{x^2}{4} \\ y=4 \end{array}\right\}$ $4=\frac{x^2}{4} \rightarrow x_1=4 \rightarrow 4=y_1$

$x_2=-4 \rightarrow y_2=4$

lehenengo koadrant $P(4,4) \rightarrow$ baina esparrutik kanpo



Bi esparru desberdinetan zatituko da.

$$A = \int_0^1 \left(4 - \frac{x^2}{4}\right) dx + \int_1^2 \left(\frac{4}{x^2} - \frac{x^2}{4}\right) dx$$

$$= \left[4x - \frac{x^3}{12}\right]_0^1 + \left[4 \cdot \frac{x^{-2+1}}{-2+1} - \frac{x^3}{12}\right]_1^2 =$$

Barrowen erregela aplikatuz

$$A = \left[4x - \frac{x^3}{12}\right]_0^1 + \left[\frac{-4}{x} - \frac{x^3}{12}\right]_1^2 =$$

$$= 4 - \frac{1}{12} - 0 + \left(\frac{-4}{2} - \frac{2^3}{12}\right) - \left(\frac{-4}{1} - \frac{1}{12}\right) =$$

$$= 4 - \cancel{\frac{1}{12}} - 2 - \frac{8}{12} + 4 + \cancel{\frac{1}{12}} = 6 - \frac{8}{12} = \frac{16}{3}$$

$$= \underline{\underline{5{,}33}}$$